SpectrumTune: Transfer Learning for Music Genre Classification

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Abstract

When streaming music, we take for granted that the music comes with genre and other informational tags which are quite accurate. These can be manually tagged by experts, or identified using Deep Learning. We use spectrograms generated from the audio of the music to see if these can be used to classify the tracks into the correct genres. We employ transfer learning techniques to pretrained models to see if the resulting models are better than training from scratch. We find that using pretrained weights is quite effective, especially when we allow pretrained models to be used as the starting point, and unfreeze more layers, which gave a test accuracy of 76%. The second best result came from adding more layers to pretrained architectures, which gave a test accuracy of 66%. Third best was starting from scratch, which gave us around 20% accuracy.

1. Introduction

Music genre classification is a fundamental task in music information retrieval that typically relies on manual tagging or traditional audio feature extraction. Deep learning approaches have shown promise, but training deep models from scratch requires substantial computational resources and large labeled datasets. Transfer learning offers an alternative by leveraging knowledge from models trained on different, often larger datasets.

The SpectrumTune project investigates whether convolutional neural networks (CNNs) pre-trained on image recognition tasks can effectively classify music genres when provided with spectrograms - visual representations of audio frequency content over time. Our research explores three complementary approaches:

1. **Layer Freezing Strategies:** Systematic evaluation of different layer freezing configurations in pre-trained ResNet50, measuring the impact on accuracy and training efficiency

2. **Architecture Extension:** Adding specialized layers on top of pre-trained ResNet50 to better adapt the model to audio classification tasks

3. **Custom Model Development:** Building and training a music genre classifier from scratch as a baseline comparison

This approach builds on previous work showing that visual domain models can be surprisingly effective for sound classification tasks when fed appropriate spectrogram representations. We hypothesize that early convolutional layers in ResNet50 capture general patterns useful across domains, while later layers may require adaptation to the specific characteristics of music spectrograms. We further investigate whether specialized architectural components can enhance this transfer, and establish performance benchmarks with custom models.

2. Related Work

Transfer learning has been shown to be effective across various domains. For audio classification specifically, previous research demonstrates that visual models transferred to spectrogram-based audio tasks can outperform audio-specific architectures trained from scratch, particularly when data is limited.

Guzhov et al. [1] introduced ESResNet, demonstrating that visual domain models could perform well on environmental sound classification when fed spectrograms. Their research showed transferred models outperformed audio-specific architectures with significantly less training.

Shorten and Khoshgoftaar [2] evaluated transfer learning from pretrained CNNs (including ResNet) to music tasks and emphasized the importance of partial fine-tuning, noting that freezing early layers while fine-tuning later ones often produces optimal results.

Choi et al. [3] explored transfer learning specifically for music classification tasks and demonstrated that models pre-trained on large music datasets could be effectively transferred to smaller datasets with different label spaces, suggesting that music representation learning may transfer across different music classification tasks.

Our work extends these findings by systematically evaluating different transfer learning strategies and architecture variants with the same base model (ResNet50) to determine optimal approaches for music genre classification using the GTZAN dataset.

3. Methodology

3.1. Dataset

We used the GTZAN Genre Collection dataset, which contains 1,000 audio tracks (30 seconds each) across 10 genres: blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock. For our experiments, we used the pre-generated spectrograms provided in the dataset, resizing them to 224×224 pixels to match ResNet50's input requirements.

The dataset was split into training (70%), validation (15%), and test (15%) sets, ensuring balanced representation across genres. Data augmentation was applied to the training set through standard image transforms like minor color jittering, to improve generalization.

We have included some example spectrograms like one for Jazz and one for Rock 1

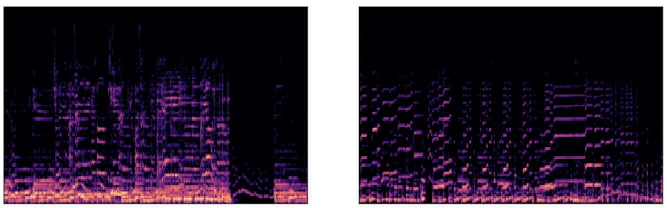

Figure 1. A sample spectrogram for a Rock music sample (left) and a Jazz music sample (right)

3.2. Research Approaches

Our team investigated three distinct but complementary approaches to music genre classification:

3.2.1 Approach 1: Layer Freezing Strategies

This approach employed a ResNet50 architecture pre-trained on ImageNet as the base model. The final fully connected layer was replaced with a new classification head consisting of:

- A linear layer mapping from the ResNet50 feature space (2048 dimensions) to 256 dimensions
- ReLU activation
- Dropout (0.3) for regularization
- A final linear layer mapping to 10 output classes (genres)

We investigated five different freezing strategies:

1. **All**: Freeze all convolutional layers, only fine-tune the classification head
2. **Layer4**: Freeze all layers except Layer4 (final convolutional block) and classification head
3. **Layer3**: Freeze all layers except Layer3, Layer4, and classification head
4. **Layer2**: Freeze all layers except Layer2, Layer3, Layer4, and classification head
5. **None**: Fine-tune the entire network (no frozen layers)

3.2.2 Approach 2: Architecture Extension

This approach extended the pre-trained ResNet50 base model with additional specialized layers:

- **Architecture 1**: Conv(2048→512) → BN → ReLU → MaxPool → Conv(512→128) → BN → ReLU → AdaptiveAvgPool(1x1) It takes an input with 2048 channels, and outputs a feature map with 512 channels. we then do some batch normalization to stabilize training, then add some non linearity before adding a maxpooling layer to reduce dimensions. We add another convolutinal layer to go down to 128 channels, after which we add another batch normalization layer and ReLU, and then an adaptive average pooling layer to bring the dimension down to 128, for our final fully connected layers for classification
- **Architecture 2**: Conv(2048→512) → BN → ReLU → Conv(512→512) → BN → ReLU → MaxPool → Conv(512→128) → BN → ReLU → AdaptiveAvgPool(1x1) Similar to Arch 1, but introduces an extra convolutional layer (512 to 512 channels) before the max-pooling layer.
- **Architecture 3**: Conv(2048→512) → BN → ReLU → Conv(512→512, dilation=2) → BN → ReLU → MaxPool → Conv(512→128) → BN → ReLU → AdaptiveAvgPool(1x1) Very similar to Architecture 2, but with a dilation in the kernel to see if that improves classification accuracy
- **Architecture 4**: Conv(2048→512) → BN → ReLU → Conv(512→256) → BN → ReLU → Conv(256→128) → BN → ReLU → AdaptiveAvgPool(1x1) Similar to Architecture 1, but removes the max pooling layer to see if that has any impact on classification accuracy

3.2.3 Approach 3: Custom Model Development

This approach involved building a specialized CNN architecture from scratch, designed specifically for music spectrogram analysis:

- CNN architecture with variable-sized filters
- 3 convolutional blocks and final Linear-ReLU-Softmax blocks
- Batch normalization and advanced regularization techniques to prevent overfitting
- Training from random initialization without transfer learning

3.3. Training Process

For all approaches, we followed a consistent training protocol:

- 20 epochs of training using the Adam optimizer with experimented learning rates ranging from of 0.0001 to 1
- Cross-entropy loss as the optimization objective
- Model saving based on best validation accuracy
- Common data pre-processing and augmentation to increase sample size

For each approach, we measured:

- Training and validation loss/accuracy during training
- Final test accuracy
- Per-genre performance metrics (precision, recall, F1-score)
- Confusion matrices to analyze misclassification patterns

4. Results and Analysis

4.1. Layer Freezing Strategies

4.1.1 Parameter Efficiency

The different freezing strategies resulted in significant variations in the number of trainable parameters, as shown in Table 1.

Table 1. Trainable parameters by freezing strategy

Strategy	Trainable Parameters	% of Total
All	527,114	2.19%
Layer4	15,491,850	64.45%
Layer3	22,590,218	93.99%
Layer2	23,809,802	99.06%
None	24,035,146	100.00%

4.1.2 Model Performance

The test accuracy results for each freezing strategy showed a clear trend, as presented in Table 2.

Table 2. Test accuracy by freezing strategy

Strategy	Test Accuracy
None	76.16%
Layer3	70.20%
Layer2	70.20%
Layer4	62.91%
All	53.64%

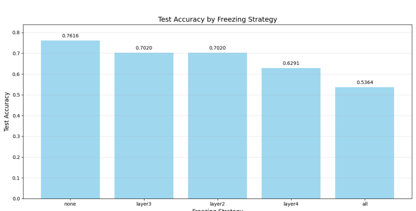

Figure 2. Test accuracy comparison across all freezing strategies, showing the progressive improvement as more layers are unfrozen. The "None" strategy achieves the highest accuracy (76.16%), while the "All" strategy performs worst (53.64%).

These results indicate that allowing more layers to be fine-tuned generally leads to better performance, with the best results achieved when fine-tuning the entire network. However, the Layer3 and Layer2 strategies achieved comparable performance while requiring fewer trainable parameters.

4.1.3 Genre-Specific Performance

Analysis of the confusion matrices revealed interesting patterns in genre classification across different freezing strategies.

Several key patterns emerged from these confusion matrices:

- **Highly Distinguishable Genres**: Jazz achieved perfect classification (100%) in the "None" strategy (Fig. 7), while metal and classical consistently showed high accuracy (¿90%) across multiple strategies.
- **Frequently Confused Genres**: Country music was commonly misclassified as rock, blues, or disco across different strategies. Similarly, rock was often confused with metal or pop, particularly in the "All" strategy (Fig. 8).
- **Progressive Improvement**: Comparing the "All" strategy (test accuracy: 53.64%, Fig. 8) with "None" (test accuracy: 76.16%, Fig. 7), we observe dramatic improvements in blues, disco (33% $\rightarrow$ 76%), and rock (20% $\rightarrow$ 67%) classification. The "All" strategy shows significantly lower diagonal values and increased misclassifications, particularly for disco (33%), country (37%), and rock (20%).
- **Challenging Genres**: Country music remained the most challenging genre to classify correctly, with only 47% accuracy even in the best-performing "None" strategy (Fig. 7). Similarly, disco proved challenging in the "All" strategy with only 33% accuracy (Fig. 8).

Table 3. Performance metrics by genre for "None" strategy

Genre	Precision	Recall	F1-Score
Blues	0.80	0.75	0.77
Classical	0.83	0.91	0.87
Country	0.69	0.47	0.56
Disco	0.64	0.76	0.70
Hip-hop	0.91	0.77	0.83
Jazz	1.00	1.00	1.00
Metal	0.85	0.92	0.88
Pop	0.74	0.82	0.78
Reggae	0.73	0.73	0.73
Rock	0.62	0.67	0.65

4.1.4 Training Dynamics

Comparing the training and validation curves across strategies revealed distinct patterns (Fig. 3). Additionally, examining the best-performing model's training behavior provided further insights:

Several key observations emerged:

1. **Convergence Speed**: More restrictive freezing strategies ("All", "Layer4") led to faster initial convergence in training but plateaued at higher loss values and lower accuracy
2. **Validation Stability**: The "All" strategy showed the most stable validation curve, while "None" exhibited higher variance, suggesting greater susceptibility to overfitting
3. **Performance Gap**: The "None" strategy had a larger gap between training accuracy (100%) and validation accuracy (85%), compared to more restrictive strategies

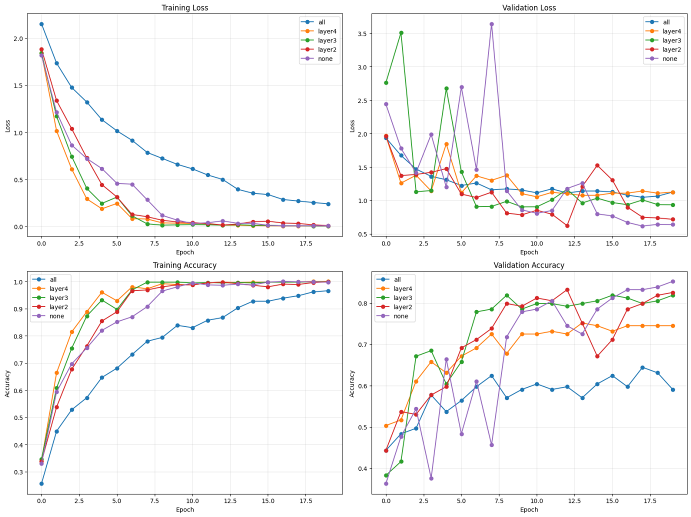

Figure 3. Training and validation metrics across all freezing strategies. Top row: Loss curves for training (left) and validation (right). Bottom row: Accuracy curves for training (left) and validation (right). Less restrictive freezing strategies show slower initial convergence but ultimately achieve better performance.

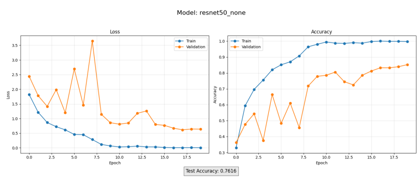

Figure 4. Training and validation metrics for the ”None” freezing strategy, showing high training accuracy (100%) and final validation accuracy (85%) with considerable fluctuation during early training. The test accuracy reached 76.16%, indicating good generalization despite the fluctuations.

4. **Optimization Difficulty**: The ”None” strategy faced the most challenging optimization landscape, as evidenced by the high variance in validation loss during early epochs (likely due to the larger number of trainable parameters)

Despite the fluctuations in the validation metrics for the ”None” strategy, it ultimately achieved the best test performance, demonstrating that complete fine-tuning was beneficial for this task, even with the increased risk of overfitting. This suggests that the model was able to learn meaningful genre-specific representations that generalized well to unseen examples.

4.2. Architecture Extension

4.2.1 Parameter Efficiency

The different Architectural extension strategies resulted in significant variations in the number of trainable parameters, which can be seen in Table 5 in the appendix.

- **Architecture 1**: Conv(2048→512) → BN → ReLU → MaxPool → Conv(512→128) → BN → ReLU → AdaptiveAvgPool(1x1) Allows us to keep some spatial invariance while reducing spatial dimensions as the layers progress

- **Architecture 2**: Conv(2048→512) → BN → ReLU → Conv(512→512) → BN → ReLU → MaxPool → Conv(512→128) → BN → ReLU → AdaptiveAvgPool(1x1) Introduces an extra convolutional layer (512 channels) before the max-pooling, which allows the model to capture more spatial information before dimension reduction

- **Architecture 3**: Conv(2048→512) → BN → ReLU → Conv(512→512, dilation=2) → BN → ReLU → MaxPool → Conv(512→128) → BN → ReLU → AdaptiveAvgPool(1x1) Very similar to Architecture 2, but with a dilation in the kernel which increases the receptive field and potentially covers more information

- **Architecture 4**: Conv(2048→512) → BN → ReLU → Conv(512→256) → BN → ReLU → Conv(256→128) → BN → ReLU → AdaptiveAvgPool(1x1) Similar to Architecture 1, but removes the max pooling layer and retains more spatial information for later layers

4.2.2 Model Performance

The test accuracy results for each extended architecture showed stiff competition between the architectures, as presented in Table 4.

Table 4. Test accuracy by Architecture, Frozen vs Fine Tuned

Strategy	Frozen- Accuracy	FineTuned- Accuracy
Architecture 1	62.25%	66.23%
Architecture 2	65.56%	65.56%
Architecture 3	64.24%	64.90%
Architecture 4	64.24%	61.59%

Because these models have a lot of common characteristics like the first unfrozen layer having 512 output dimensions for the Conv block - Batch Norm - ReLU pattern, it isn’t too surprising in retrospect that the results are quite similar.

On the individual runs, we did se a 2-4% gain in accuracy when the models were allowed to unfreeze all layers for the final 1-2 epochs.

4.2.3 Genre-Specific Performance

Analysis of the confusion matrices revealed interesting patterns in genre classification across different extending architectures.

Several key patterns emerged from these confusion matrices:

- **Highly Distinguishable Genres**: The best model with the extended architecture achieved a high classification accuracy (92%) on the Jazz and Pop categories in the fine-tuned strategy (Fig.9). Similarly, the frozen weights model also achieved near perfect classification (92%) for Jazz and Pop (Fig.10). Classical consistently showed high accuracy (¿80%) across multiple architectures.

- **Frequently Confused Genres**: Country music was commonly misclassified as pop or disco across different architectures. Similarly, rock was often confused with metal or pop for the extended architectures, very similar to the earlier freezing strategies that were discussed.

- **Progressive Improvement**: Comparing the best fine-tuned model (test accuracy: 66.24%, Fig.9) with the best frozen weights model (test accuracy: 61.59%, Fig.10), we observe improvements in most genres. The diagonal represents correct classifications with darker blue indicating higher accuracy.

- **Challenging Genres**: Country music remained the most challenging genre in the fine-tuned model with only 44% accuracy (Fig.9), while disco proved most challenging in the frozen weights model with 33% accuracy (Fig.10), which is similar to the classical and country music being difficult in the custom models.

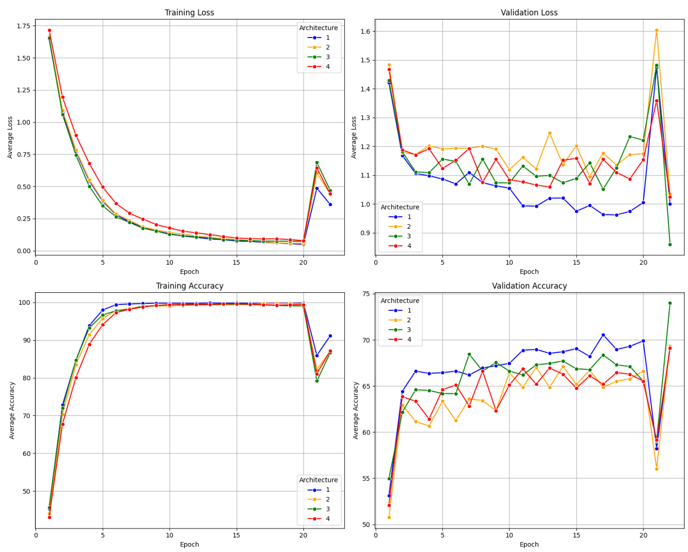

Figure 5. Training and validation metrics across all tested extended architectures. Top row: Loss curves for training (left) and validation (right). Bottom row: Accuracy curves for training (left) and validation (right). Architecture number 1 in blue has the quickest conversion and highest accuracy on the frozen rounds. We see slight jumping behavior in accuracy for the final 2 epochs, because we tried out fine tuning on the entire set of weights which resulted in better results for some architectures

4.2.4 Training Dynamics

Comparing the training and validation curves across the various architectures revealed distinct patterns (Fig. 5). Additionally, examining the best-performing model's training behavior provided further insights:

Several key observations emerged:

1. **Convergence Speed**: Most architectures converted at about the same rate, mostly because they involved similar layer types, number of parameters. Architecture 3 converged the quickest by a little bit, which might be the result of the dilation, which helps capture features at different scales in the image

2. **Validation Stability**: Architecture 1 showed the most stable validation curve, with archtecture 4 showing the most variation in validation curve.

3. **Performance Gap**: There was a major gap in training ves validation accuracy. Near 100% for training vs around 65-70% for validation gives an indication that the sample sizes were not enough, and might be resulting in overfitting on the limited data.

4.3. Custom Model Development

4.3.1 Model Performance

The test accuracy results for our experimented custom architecture showed an accuracy of 20% on our test set.

4.3.2 Genre-Specific Performance

Analysis of the confusion matrices revealed interesting patterns in genre classification across different extending architectures. (Fig. 11)

The following observations were made on this confusion matrix: The following observations were made on this confusion matrix:

- **Best Distinguishable Genres**: The custom architecture achieved the highest classification accuracy on Jazz and Metal categories (33% each) (Fig.11). Pop was also relatively well categorized at 22%. The diagonal represents correct classifications with darker blue indicating higher accuracy.

- **Most predicted Genres**: In many observations, the model liked to predict Reggae, Hip-hop, Jazz or Pop more than other categories (Fig.11).

- **Challenging Genres**: Classical and Country music proved most challenging with 0% accuracy each (Fig. 11), almost worse than chance. This represents the most significant classification challenges for the custom model with a test accuracy of 20%.

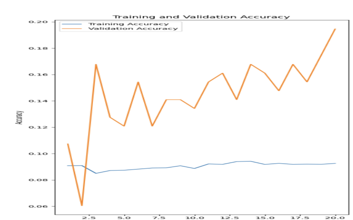

Figure 6. Training and validation metrics for the custom architecture.

4.3.3 Training Dynamics

Looking at the training and validation curve (Fig. 6) for the custom architecture showed that it doesn't have the sharp then gradual increase in accuracy which we expect to be aspects of a good model. This gives an indication that even though this was the best model from the custom model hyperparameter tuning, it doesn't do nearly as well as the other strategies. The validation and text accuracy were never high and stayed between 6% and 20% accuracy, which is very close to random guessing.

5. Discussion

5.1. Layer Freezing Insights

Our findings on layer freezing strategies reveal that visual features learned from ImageNet are indeed transferable to music genre classification via spectrograms, but the optimal transfer strategy involves allowing significant adaptation of the model to the new domain.

The performance improvements seen when unfreezing more layers indicate that while early convolutional features may capture some relevant patterns, the higher-level representations encoded in later layers require substantial adaptation to effectively represent music genre characteristics. This aligns with the intuition that low-level visual features (edges, textures) might have some correspondence with spectrogram patterns, but higher-level features optimized for object recognition would require significant retraining to identify genre-specific patterns.

The considerable performance difference between "All" (53.64%) and "None" (76.16%) freezing strategies - a 22.52% absolute improvement - demonstrates that simply using ImageNet features with a new classification head is insufficient for optimal music genre classification. The progressive improvement observed as more layers are unfrozen suggests a hierarchical transfer of relevance:

- **Early layers** (captured in "All" strategy): Basic visual features transfer reasonably well to spectrograms, enabling above-random classification
- **Middle layers** (added in "Layer4" and "Layer3" strategies): Mid-level representations require adaptation but contribute substantially to improved accuracy (+16.56% from "All" to "Layer3")
- **Final layers** (added in "None" strategy): Complete fine-tuning offers further benefits (+5.96% from "Layer3" to "None") but with diminishing returns relative to computational cost

The comparable performance between Layer2/Layer3 freezing and complete fine-tuning (None) suggests a potential compromise between computational efficiency and model performance. For applications with limited computational resources, freezing early layers may offer a reasonable trade-off, achieving 92% of the performance of full fine-tuning while training only 94% of the parameters.

5.2. Architecture Extensions Insights

Our findings on architecture extension help us learn that visual features learned from ImageNet are indeed transferable to music genre classification via spectrograms, but the optimal transfer strategy involves allowing significant adaptation of the model to the new domain.

The performance improvements seen when adding more layers indicates that while early convolutional features from the original ResNet may capture some relevant patterns, the higher-level representations encoded in the added layers require a decent bit of adaptation and training to effectively represent music genre characteristics. This aligns with the earlier insight that low-level visual features (edges, textures) might have some correspondence with spectrogram patterns, but higher-level features optimized for object recognition would require significant retraining to identify genre-specific patterns.

The considerable performance difference between "All" from the frozen weights strategies of the previous section (53.64%) and the best extended architecture model (66.23%) - an 12.59% absolute improvement - demonstrates that simply using ImageNet features with a new classification head is insufficient for optimal music genre classification.

We also learned that on average, by letting all the weights be unfrozen for two final iterations, we would see an accuracy gain of 3%, which suggests that there are other combinations and architectures that can potentially do better if explored

5.3. Custom Model Insights

Our findings on trying a custom model show us that Other architecture variations of this model were tried, but none did any better. This indicated that these architectures that used convolution, max pooling, batch normalization, ReLU, linear, and softmax functions or layers were not enough. That could mean that more functions or layers need to be used or other kinds of them need to be used like the ResNet models. These models had more layers indicating that a simpler model would not do a good job. Training budget constraints limited us in trying more architectures for this option, which limits our ability to make comments. What was abundantly clear, was that our model was underfitting and that we needed more convolutional blocks to get closer to more reasonable accuracy and F1 scores.

6. Conclusion and Future Work

Our experiments with layer freezing strategies for ResNet50 on the GTZAN dataset yield several important insights for transfer learning in music genre classification.

The results demonstrate that knowledge from visual domains can effectively transfer to audio classification tasks, with the best performance achieved through full fine-tuning of the pre-trained model (76.16% accuracy). However, selective freezing of early layers offers a compelling trade-off between performance and computational efficiency, with Layer3 freezing achieving 70.20% accuracy while training only 94% of the parameters.

The progressive improvement observed as more layers are unfrozen reveals that while low-level visual features transfer reasonably well to spectrogram patterns, higher-level representations require substantial adaptation to the audio domain. This suggests that future transfer learning approaches should focus on efficient ways to adapt these higher-level features while preserving the general pattern recognition capabilities of early layers.

The genre-specific analysis reveals consistent strengths in classifying classical, metal, and jazz genres across all freezing strategies, while rock and country prove more challenging. This indicates that certain musical characteristics may be more readily identified in spectrograms regardless of model adaptation, while others require more extensive fine-tuning.

Future work on layer freezing could explore:

1. More granular freezing strategies, selectively unfreezing specific blocks within layers
2. Progressive unfreezing during training to potentially achieve better convergence
3. Analysis of attention patterns and feature activations to better understand which visual features transfer well to audio
4. Evaluation on larger and more diverse music datasets to test generalizability

Future work on architectural extension for transfer learning could explore:

1. Adding more convolutional blocks to see which ones help

2. trying more fine tuning to the models and unfreeze the entire architecture for more iterations

3. using semi supervised learning to augment the dataset to get richer training data and hence more robust results

Overall, our findings contribute to the growing body of evidence supporting cross-domain transfer learning and provide practical guidelines for implementing efficient music classification systems using deep learning.

References

[1] A. Guzhov, F. Raue, J. Hees, and A. Dengel, "ESResNet: Environmental Sound Classification Based on Visual Domain Models," arXiv:2004.07301, 2021. 1

[2] C. Shorten and T. M. Khoshgoftaar, "A survey on Image Data Augmentation for Deep Learning," Journal of Big Data, vol. 6, no. 1, pp. 60, 2019. 1

[3] K. Choi, G. Fazekas, M. Sandler, and K. Cho, "Transfer Learning for Music Classification and Regression Tasks," arXiv:1703.09179, 2017. 1

[4] G. Tzanetakis and P. Cook, "Musical genre classification of audio signals," IEEE Transactions on Speech and Audio Processing, vol. 10, no. 5, pp. 293-302, 2002.

[5] "ImageFilter Module." Pillow (PIL Fork), pillow.readthedocs.io/en/stable/reference/ImageFilter.html. Accessed 28 Apr. 2025.

A. Additional Figures

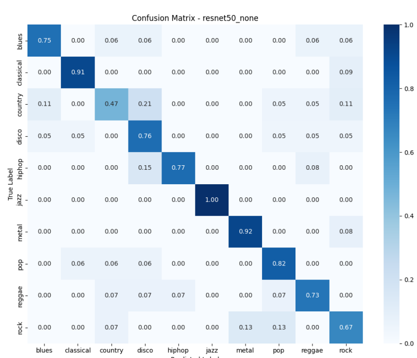

Figure 7. Confusion matrix for "None" freezing strategy (test accuracy: 76.16%). The diagonal represents correct classifications, with darker blue indicating higher accuracy. Jazz achieved perfect classification (100%), while country music proved most challenging (47%).

Table 5. Trainable parameters by selected architecture

Strategy	Trainable Parameters
Architecture 1	10,030,218
Architecture 2	12,391,050
Architecture 3	12,391,050
Architecture 4	10,915,722

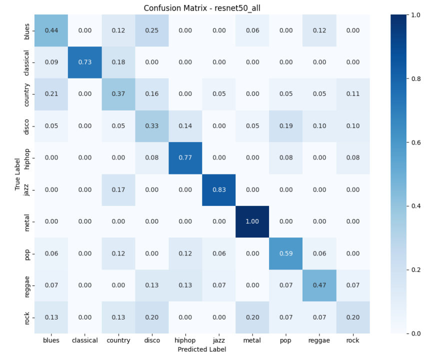

Figure 8. Confusion matrix for "All" freezing strategy (test accuracy: 53.64%). Note the significantly lower diagonal values and increased misclassifications compared to the "None" strategy, particularly for disco (33%), country (37%), and rock (20%).

Figure 9. Confusion matrix for the best fine tuned model in the Extended Architecture Strategy(test accuracy: 66.24%). The diagonal represents correct classifications, with darker blue indicating higher accuracy. Jazz and Pop achieved near perfect classification (92%), while country music proved most challenging (44%).

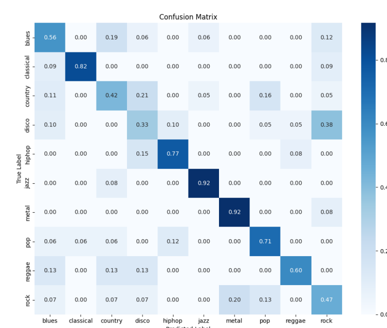

Figure 10. Confusion matrix for the best frozen weights model in the Extended Architecture Strategy(test accuracy: 61.59%). The diagonal represents correct classifications, with darker blue indicating higher accuracy. Jazz and Pop achieved near perfect classification (92%), while disco music proved most challenging (33%).

B. Hyperparameter Tuning Details- Custom Model

The model trained on a dataset augmented to 32 variations per image (seven filters—blur, Gaussian blur, median, mode, smooth, sharpen, unsharpmask—applied to three crops). Eighteen hyperparameter combinations (batch sizes 32/64, regularization 1/10/100, learning rates 0.0001/0.01/1) were tested on a CNN with stacked 3×3 and 2×2 convolutional layers, max pooling, batch normalization, ReLU, a linear layer, and softmax. The highest validation accuracy reached only 19.46% (batch 32, regularization 10, LR 0.0001), indicating suboptimal performance

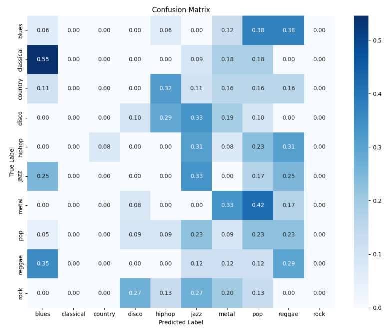

Figure 11. Confusion matrix for the best custom model (test accuracy: 20%). The diagonal represents correct classifications, with darker blue indicating higher accuracy. Jazz, Metal and Pop were the best categorized (33%, 33% and 22% respectively), while classical and country music proved most challenging (0% each).

and a need for architectural refinement or broader hyperparameter tuning.

Table 6. Custom Architecture- Hyperparameter Tuning

Batch Size	Regularization	Learning Rate	Final Training Loss	Final Training Accuracy	Final Validation Loss	Final Validation Accuracy	Test Loss	Test Accuracy
32	10	0.0001	73.68303343	0.092542918	2.300229154	0.194630872	2.302405661	0.119205298
32	100	0.0001	73.68354044	0.098444206	2.300858734	0.127516779	2.302448003	0.072847682
64	1	0.0001	73.68263908	0.098310086	2.302383578	0.127516779	2.302153047	0.132450331
32	1	0.01	73.68271696	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
32	10	0.01	73.68271696	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
32	100	0.01	73.68271696	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
32	1	1	73.68271696	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
32	10	1	73.68271696	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
32	100	1	73.68271696	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
64	10	0.0001	73.68253587	0.095940629	2.2993966	0.10738255	2.300567551	0.092715232
64	1	0.01	73.68264523	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
64	10	0.01	73.68264523	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
64	100	0.01	73.68264523	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
64	1	1	73.68264523	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
64	10	1	73.68264523	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
64	100	1	73.68264523	0.097281831	2.302584954	0.10738255	2.302584959	0.105960265
64	100	0.0001	73.68284616	0.097281831	2.303295624	0.073825503	2.305988642	0.112582781
32	1	0.0001	73.68335473	0.093124106	2.306122343	0.067114094	2.301551522	0.059602649

C. Team Contributions

Team Member	Contribution
Abhishek Singh	Abhishek laid the groundwork for the project by setting up the base layer implementation and establishing the GitHub repository. He created the initial LaTeX report template and focused on the layer freezing experiments, implementing the five different freezing strategies for the ResNet50 model. His work included training these models and analyzing their performance metrics, parameter efficiency, and genre-specific classification patterns.
Umair Mesiya	Umair led the exploratory data analysis at the beginning of the project and was responsible for the model architecture extension experiments. He designed and tested four different architectural variations that build upon the pretrained ResNet50, evaluating both frozen and fine-tuned versions of each architecture. He also contributed to refining the final report structure and formatting based on the LaTeX template requirements.
Nicholas Jerdack	Nicholas worked on the custom CNN model implementation, designing the architecture from scratch with variable kernel sizes and specialized regularization techniques. He was also responsible for implementing the data augmentation pipeline, which helped improve model generalization. Additionally, he conducted extensive hyperparameter tuning experiments and performed comparative analysis between the custom model and transfer learning approaches.

Table 7. Team Contributions Summary